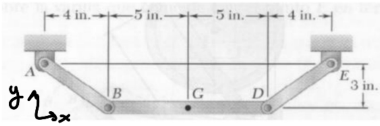
CCR

TP1

1. Si en el instante mostrado la barra AB tiene velocidad angular constante de 15 rad/s en sentido contrario al de las manecillas del reloj, determine a) la aceleración angular de la barra DE, b) la aceleración del punto G.



$$V_B = V_A + W_{AB} \times (\overline{B-A}) = 45\check{i} + 60\check{j}$$

$$\begin{vmatrix} 0 & 0 & 15 \\ 4 & -3 & 0 \end{vmatrix} = (45; 60; 0)$$

$$V_D = V_E + W_{DE} \times (\overline{D-E}) \rightarrow V_D = -3W_{DE}\,\check{i} + 4W_{DE}\check{j}$$

$$\begin{vmatrix} 0 & 0 & W_{DE} \\ -4 & -3 & 0 \end{vmatrix} = (3W_{DE}; -4W_{DE}; 0)$$

$$V_D = V_B + W_{BD} \times (\overline{D-B}) \rightarrow V_D = 45\check{i} + 60\check{j} + 10W_{BD}\check{j}$$

$$\begin{vmatrix} 0 & 0 & W_{BD} \\ 10 & 0 & 0 \end{vmatrix} = (0; 10W_{BD}; 0)$$

$3W_{DE} = 45 \rightarrow W_{DE} = 15$

$-4W_{DE} = 60 + 10W_{BD}$

$W_{BD} = -12\,s^{-1}$

$V_D = -45\check{i} + 60\check{j}$

- Calculo aceleraciones

$$a_B = \cancel{\overset{\nearrow w \text{ es cte}}{\gamma_{AB} \times (\overline{B-A})}} + W_{AB} \times \left[W_{AB} \times (\overline{B-A})\right] \rightarrow a_B = -900\check{i} + 675\check{j}$$

$$(*) \begin{vmatrix} 0 & 0 & 15 \\ 45 & 60 & 0 \end{vmatrix} = (-900; 675; 0)$$

$$a_D = a_B + \underbrace{\gamma_{BD} \times (\overline{D-B})}_{\mathrm{I}} + W_{BD} \times \underbrace{\left[W_{BD} \times (\overline{D-B})\right]}_{\mathrm{II}} \rightarrow a_D = -900\check{i} + 675\check{j} + 10\gamma_{BD}\check{j} - 1440\check{i}$$

$$\mathrm{I} \begin{vmatrix} 0 & 0 & \gamma_{BD} \\ 10 & 0 & 0 \end{vmatrix} = 10\gamma_{BD}\check{j} \qquad \mathrm{II} \begin{vmatrix} 0 & 0 & -12 \\ 0 & -120 & 0 \end{vmatrix} = -1440$$

$a_D = \cancel{a_E} + \gamma_{DE} \times (\overline{D-E}) + \omega_{DE} \times \left[\omega_{DE} \times \left[\overline{E-D}\right]\right] \Rightarrow a_D = 3\gamma_{DE}\,\hat{i} - 4\gamma_{DE}\,\hat{j} + 900\,\hat{i} + 675\,\hat{j}$

I $\qquad$ II

$a_D = -2340\,\hat{i} + 4995\,\hat{j}$

I $\begin{vmatrix} 0 & 0 & \gamma_{DE} \\ -4 & -3 & 0 \end{vmatrix} = (3\gamma_{DE}; -4\gamma_{DE}; 0)$ $\qquad$ II $\begin{vmatrix} 0 & 0 & 15 \\ 45 & -60 & 0 \end{vmatrix} = (900; 675; 0)$

Igualando en $\hat{i}$: $\quad -900 - 1440 = 3\gamma_{DE} + 900 \rightarrow \gamma_{DE} = -1080$

$\hat{j}$: $\quad 675 + 10\gamma_{BD} = -4\gamma_{DE} + 675 \rightarrow \gamma_{BD} = 432$

$a_G = a_D + \gamma_{BD} \times \left[\overline{G-D}\right] + \omega_{BD} \times \left[\omega_{BD} \times \left[\overline{G-D}\right]\right] \rightarrow -2340\,\hat{i} + 4995\,\hat{j} - 2160\,\hat{j} + 720\,\hat{i}$

I

$a_G = -1620\,\hat{i} + 2835\,\hat{j}$

I $\begin{vmatrix} 0 & 0 & 432 \\ -5 & 0 & 0 \end{vmatrix} = (0; -2160; 0)$

II $\begin{vmatrix} 0 & 0 & -12 \\ -5 & 0 & 0 \end{vmatrix} = (0; +60; 0)$ $\qquad \begin{vmatrix} 0 & 0 & -12 \\ 0 & +60 & 0 \end{vmatrix} = (+720; 0; 0)$

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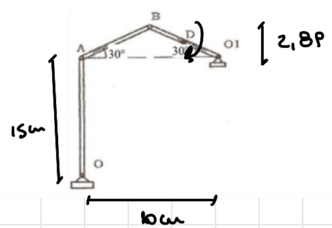
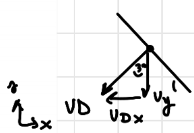
2. Determinar las velocidades angulares de las barras correspondientes al mecanismo compuesto por las barras OA; AB yBO1 que se mueven en el plano del dibujo. Considerar que la velocidad del punto medio D de la barra BO1 tiene, para el instante indicado un valor de 25 cm/s, y gira en el sentido de las agujas del reloj, siendo OA = 15 cm yAO1 = 10 cm. (ángulo barras AB, BO1 = 30°)



$V_D = 25\ cm/s$

$V_{Dx} = V_D \cdot sen(30) = 12,5$

$V_{Dy} = V_D \cdot cos(30) = 21,65$



$V_B = V_D + \omega_{BO1} \times (\overline{B-D}) \rightarrow V_B = 12,5\,\hat{i} + 21,65\,\hat{j} - 1,45\,\omega_{BO1}\,\hat{i} - 2,5\,\omega_{BO1}\,\hat{j}$

$\begin{vmatrix} 0 & 0 & \omega_{BO1} \\ -2,5 & 1,45 & 0 \end{vmatrix} = (-1,45\,\omega_{BO1}; -2,5\,\omega_{BO1}; 0)$

$V_B = \cancel{V_{O1}} + \omega_{BO1} \times (\overline{B-O_1}) \rightarrow V_B = -2,89\,\omega_{BO1}\,\hat{i} - 5\,\omega_{BO1}\,\hat{j}$

$\begin{vmatrix} 0 & 0 & \omega_{BO1} \\ -5 & 2,89 & 0 \end{vmatrix} = (-2,89\,\omega_B; -5\,\omega_{BO1}; 0)$

$$-2{,}89\,\omega_{BO1} = -14{,}5\,\omega_{BO1} + 12{,}5 \rightarrow \omega_{BO1} = -8{,}68 \qquad v_B = 25\,\hat{i} + 43{,}4\,\hat{j}$$

$$V_A = V_B + \omega_{AB} \times (\overline{A-B}) \rightarrow 25\,\hat{i} + 43{,}4\,\hat{j} + 2{,}89\,\omega_{AB}\,\hat{i} - 5\,\omega_{AB}\,\hat{j}$$

$$\begin{vmatrix} 0 & 0 & \omega_{AB} \\ -5 & -2{,}89 & 0 \end{vmatrix} = (2{,}89\,\omega_{AB};\ -5\,\omega_{AB};\ 0)$$

$$V_A = \cancel{V_O} + \omega_{AO} \times (\overline{A-O}) \rightarrow -15\,\omega_{AO}\,\hat{i}$$

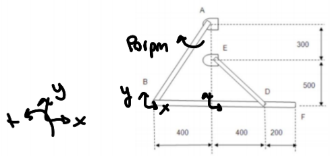
$$\begin{vmatrix} 0 & 0 & \omega_{AO} \\ 0 & 15 & 0 \end{vmatrix} = (-15\,\omega_{AO};\ 0;\ 0)$$

$+43{,}4 - 5\,\omega_{AB} = 0 \rightarrow \omega_{AB} = 8{,}68\ s^{-1}$

$-15\,\omega_{AO} = 25 + 2{,}89\,\omega_{AB}$

$\omega_{AO} = -3{,}33$

(3) **3.** El mecanismo de la figura está compuesto de las barras AB, ED y BDF. Las dos primeras giran alrededor de los puntos A y E respectivamente. Si la barra AB gira con una velocidad angular ωAB= 90 rpm en sentido horario, calcular las velocidades angulares vectoriales de las barra BDF y ED, y la velocidad vectorial del punto F. (Medidas en mm) El mecanismo es encuentra articulado en A, B, D y E.



$1\ rpm = 2\pi/60$

$90\ rpm = 9{,}42\ s^{-1}$

$$V_B = \cancel{V_A} + \omega_{AB} \times (\overline{B-A})$$

$$\begin{vmatrix} 0 & 0 & -9{,}42 \\ -400 & -800 & 0 \end{vmatrix} = (-7536;\ +3768;\ 0) \rightarrow V_B = -7536\,\hat{i} + 3768\,\hat{j}$$

$$V_F = V_B + \omega_{BDF} \times (\overline{F-B}) \rightarrow V_F = -7536\,\hat{i} + 3768\,\hat{j} + 1000\,\omega_{BF}\,\hat{j} \rightarrow V_F = -7536\,\hat{i} - 8442\,\hat{j}$$

$$\begin{vmatrix} 0 & 0 & \omega_{BF} \\ 1000 & 0 & 0 \end{vmatrix} = (0;\ 1000\,\omega_{BF};\ 0)$$

$$V_D = V_B + \omega_{BDF} \times (\overline{D-B}) \rightarrow V_D = -7536\,\hat{i} + 3768\,\hat{j} + 800\,\omega_{BF}\,\hat{j}$$

$$\begin{vmatrix} 0 & 0 & \omega_{BF} \\ 800 & 0 & 0 \end{vmatrix} = (0;\ 800\,\omega_{BF};\ 0)$$

$-19060\,\hat{i}$

$-7536 = 500\,\omega_{DE}$

$\omega_{DE} = -15$

$V_D = -7500\,\hat{i} - 6000\,\hat{j}$

$\omega_{BF} = -12{,}21$

$$V_D = \cancel{V_E} + \omega_{DE} \times (\overline{D-E}) \rightarrow V_D = 500\,\omega_{DE}\,\hat{i} + 400\,\omega_{DE}\,\hat{j}$$

$$\begin{vmatrix} 0 & 0 & \omega_{DE} \\ 400 & -500 & 0 \end{vmatrix} = (500\,\omega_{DE};\ 400\,\omega_{DE};\ 0)$$

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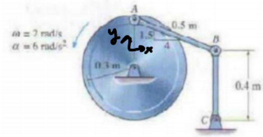
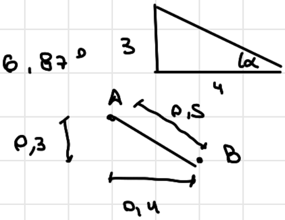
4. Para el mecanismo de la figura, la velocidad y la aceleración angular del volante es $\omega_{AB} = 2$ rad/s y $\alpha_{AB} = 6$ rad/s² respectivamente. Determine la aceleración angular de las barras AB y BC en este instante. La pendiente de la línea AB, respecto de la horizontal, es de 3/4.



$W_{AB} = 2$ rad/s

$\alpha_{AB} = 6$ rad/s²

$tg^{-1}\left(\frac{3}{4}\right) = 36{,}87°$



$V_A = \cancel{V_O} + W_{AO} \times (\overline{A-O}) \rightarrow \boxed{V_A = -0{,}6\,\hat{i}}$

$$\begin{vmatrix} 0 & 0 & 2 \\ 0 & 0{,}3 & 0 \end{vmatrix} = -0{,}6\,\hat{i}$$

$V_B = V_A + W_{AB} \times (\overline{B-A}) \rightarrow V_B = -0{,}6\,\hat{i} + 0{,}3\,W_{AB}\,\hat{i} + 0{,}4\,W_{AB}\,\hat{j}$

$$\begin{vmatrix} 0 & 0 & W_{AB} \\ 0{,}4 & -0{,}3 & 0 \end{vmatrix} = (0{,}3\,W_{AB};\ 0{,}4\,W_{AB};\ 0)$$

$V_B = \cancel{V_C} + W_{BC} \times (\overline{B-C}) \rightarrow \boxed{V_B = -0{,}4\,W_{BC}\,\hat{i}} \rightarrow \boxed{V_B = -0{,}6 \text{ m/s}\ \hat{i}}$

$$\begin{vmatrix} 0 & 0 & W_{BC} \\ 0 & 0{,}4 & 0 \end{vmatrix} = (-0{,}4\,W_{BC};\ 0;\ 0)$$

$V_B = V_B \rightarrow$

$0{,}4\,W_{AB} = 0 \rightarrow \boxed{W_{AB} = 0}$

$-0{,}6 = -0{,}4\,W_{BC} \rightarrow \boxed{W_{BC} = 1{,}5\ s^{-1}}$

$a_A = \cancel{a_O} + \underbrace{\gamma_{AO} \times (\overline{A-O})}_{I} + W_{AO} \times [\underbrace{W_{AO} \times (\overline{A-O})}_{-0{,}6\,\hat{i}}]_{II} = \boxed{-1{,}8\,\hat{i} - 1{,}2\,\hat{j} = a_A}$

$$\begin{vmatrix} 0 & 0 & 6 \\ 0 & 0{,}3 & 0 \end{vmatrix} = (-1{,}8;\ 0;\ 0)$$

$$\begin{vmatrix} 0 & 0 & 2 \\ -0{,}6 & 0 & 0 \end{vmatrix} = (0;\ -1{,}2;\ 0)$$

$a_B = a_A + \gamma_{AB} \times (\overline{B-A}) + W_{AB} \times \overset{0}{\cancel{[W_{AB} \times (\overline{B-A})]}} \rightarrow \boxed{a_B = -1{,}8\,\hat{i} - 1{,}2\,\hat{j} + 0{,}3\,\gamma_{AB}\,\hat{i} + 0{,}4\,\gamma_{AB}\,\hat{j}}$

$$\begin{vmatrix} 0 & 0 & \gamma_{AB} \\ 0{,}4 & -0{,}3 & 0 \end{vmatrix} = (0{,}3\,\gamma_{AB};\ 0{,}4\,\gamma_{AB};\ 0)$$

$a_B = -1{,}57\,\hat{i} - 0{,}8\,\hat{j}$

$$a_B = \cancel{a_C} + \underbrace{\gamma_{BC} \times (\overline{B-C})}_{I} + \omega_{BC} \times \left[\omega_{BC} \times (\overline{B-C})\right] \rightarrow a_B = -0,4\,\gamma_{BC}\,\check{i} - 0,9\,\check{j}$$

$$\begin{vmatrix} 0 & 0 & \gamma_{BC} \\ 0 & 0,4 & 0 \end{vmatrix} = (-0,4\,\gamma_{BC}; 0; 0)$$

$$\begin{vmatrix} 0 & 0 & 1,5 \\ -0,6 & 0 & 0 \end{vmatrix} = (0; -0,9; 0)$$

$$-0,9 = -1,2 + 0,4\,\gamma_{AB}\,\check{j}$$

$$\gamma_{AB} = 0,75$$

$$-1,8 + 0,3 \cdot 0,75 = -0,4\,\gamma_{BC}$$

$$\gamma_{BC} = 3,93$$